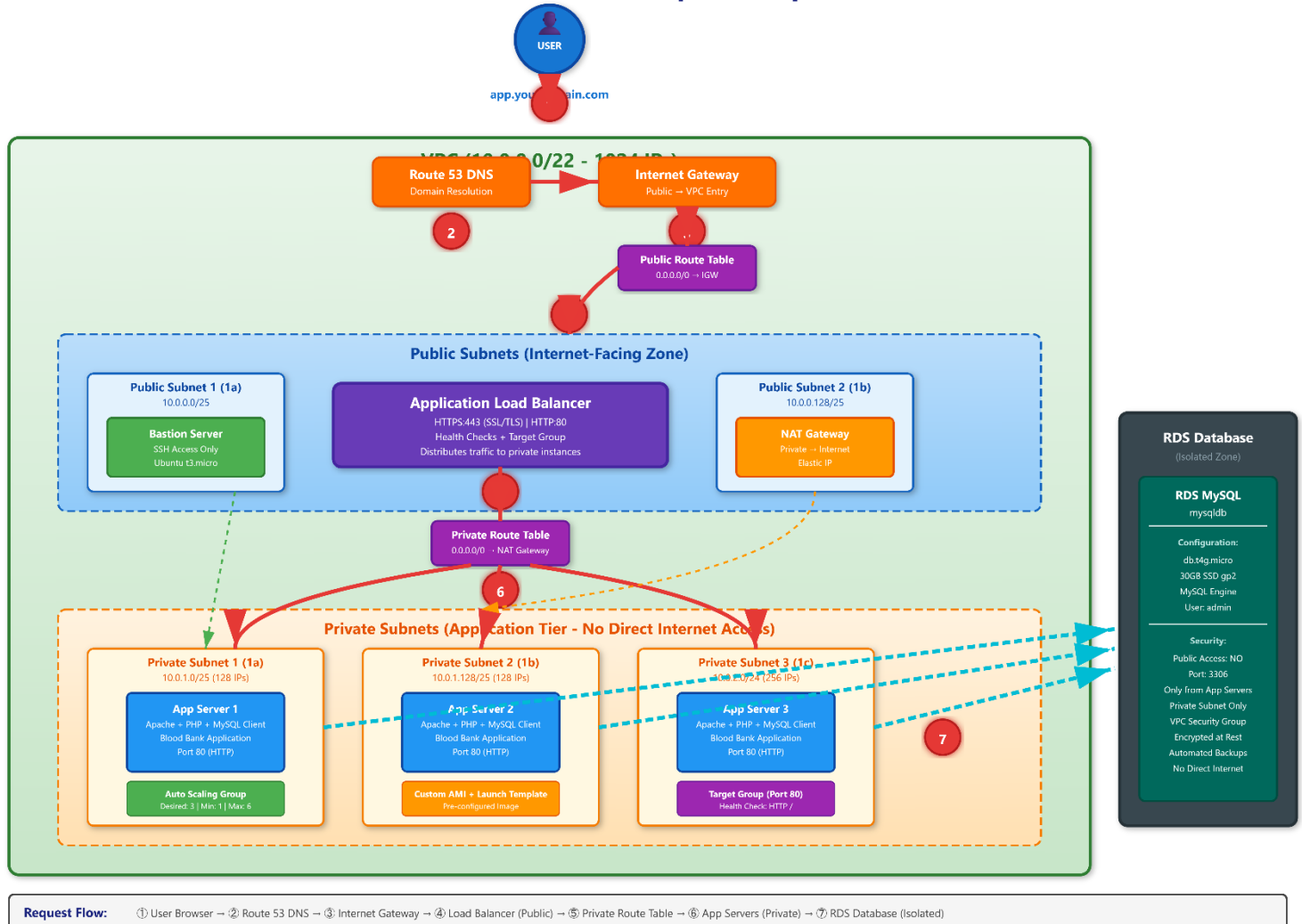


AWS CLOUD DEPLOYMENT PROJECT – BLOODBANKPROJECT

- GIT LINK: <https://github.com/AKHILESHRAVURI2001/BloodBankDonationWebsite.git> (bloodBankCode)

AWS Blood Bank - Complete Request Flow



PLAN:

- First, we want to create the INFRA (**NETWORKING PART**)
 - o VPC
 - o 3 Private Subnets
 - o 2 public Subnets
 - o NAT Gateway (used for connection for public-private)
 - o Route tables (for every subnet)
 - o Internet Gateway (attach to public route table)
- Create the EC2 in public and check and create the AMI and attach to Auto Scaling group and create in private subnet
- Add the Load Balancer in the public subnet to divide the request

PROCEDURE:

- **Create NETWORKING**
 - o First Create VPC
 - IPv4 CIDR - **10.0.0.0/22**
 - o Edit VPC
 - Edit Settings – Enable DNS hostnames

- Create Subnets
 - Create Public Subnets (2) with required IPS (128,128)
 - Select Your VPC
 - Select Availability Zone (1a,1b)
 - Give Required Subnets - **10.0.0.0/25; 10.0.0.128/25**
 - Create Private Subnets (3) with required IPS(128,128,256)
 - Select Your VPC
 - Select Availability Zone (1a,1b,1c)
 - Give Required Subnets - 10.0.1.0/25; 10.0.1.128/25; 10.0.2.0/24
- Once Create Subnets
 - Edit public subnets
 - Enable auto-assign public IPv4 address (To make as Public)
- Create Route tables
 - Create Public Route Table
 - Already Created When created VPC(Rename)
 - Create Private Route Table
 - Select VPC and Create
- Create Internet Gateway
 - Create with Required Name
 - Once Create Attach to VPC
- Create NAT GATEWAY
 - Give Name
 - Select VPC
 - Availability Zone – Zonal
 - Select Subnet(public)
 - Give Elastic IP
 - Allocate
- Attach Internet Gateway & Subnet Associate
 - Go To Public Route Tables & Edit Route
 - Add the Allow Everyone (0.0.0.0/0) and Internet Gateway
 - Go to Subnet Association and select Public subnets
- Attach NAT Gateway & Subnet Associate
 - Go To Private Route Tables & Edit Route
 - Add the Allow Everyone(0.0.0.0/0) and NAT Gateway
 - Go to Subnet Association and select Private subnets
- Create Security Group
 - Give Name
 - Select VPC
 - Add Inbound rules (All Traffic IPV4)
 - Add Inbound rules (SSH – 22 -All Traffic)
 - Add Outbound rules (All Traffic IPV4)
- **Create RDS**
 - Fully configure
 - MYSQL
 - Free Tier
 - DB Instance – mysqlpdb
 - Username : admin
 - Password: admin123
 - Storage: SSD gp2 with 30Gb
 - Size: db.t4g.micro
 - Select VPC
 - Public Access: No
 - Select Created Security Group
- **Create EC2 Instance**
 - Instance1(Basin Server)
 - Ubuntu

- T3micro
 - Keypair
 - Created VPC select
 - Public subnet (basin server)
 - Created security group
 - **8gb**
- Instance2(App Server)
 - Ubuntu
 - T3micro
 - Keypair
 - Created vpc
 - Private subnet
 - Created sg
 - 8gb
- Connection for Private Server to Public
 - Connect Basin server in the public give the ssh of the private(app) server. U will get connect by keeping pem file inside
 - Go to Root in APPSERVER(PRIVATE)
 - Install git and get the code
 - **sudo apt update**
 - **sudo apt install git**
 - **git --version**
 - get the git repo project and check the configuration changes like db
 - Install httpd and start
 - **sudo apt install apache2**
 - **sudo systemctl start apache2**
 - Install PHP, PHP with Apache, etc. dependencies
 - **sudo apt install php libapache2-mod-php php-mysql php-cli php-curl php-gd php-mbstring php-xml php-zip -y**
 - **sudo a2enmod php8.1**
 - Install MYSQL
 - **sudo apt install mysql-server**
 - **sudo systemctl start mysql**
 - Check the ur end point of mysql is working or not
 - **mysql -h <endpoint> -u admin -p**
 - Now Create DB and schema with
 - **GRANT ALL PRIVILEGES ON mydb.* TO 'admin'@'%' WITH GRANT OPTION;**
 - Move all ur repo code to
 - **/var/www/html/**
 - **sudo chown -R www-data:www-data /var/www/html/**
 - **sudo chmod -R 755 /var/www/html/**
 - **VERIFY APPSERVER Working Fine or not?**
 - Create Target Group
 - Name
 - Select OUR VPC
 - SELECT APP SERVER
 - Include Register Target 80
 - Create Test Load Balancer and Verify
 - Application Load Balancer
 - VPC
 - Subnets(public)
 - Security Group
 - Target Group
 - **APP CAN BE ACCESSBLE SUCCESSFULLY**
 - **CREATE AMI(IMAGE)**
 - Select APPSERVER and create IMAGE

- Give name and description
 - By default snapshot will create for the EBS volume
- **CREATE LAUNCH TEMPLATE**
 - Give Name & Description
 - Select your AMI
 - Select upcoming creating details created from AMI
 - Instance Type, Key Pair, Subnet(public/private), security Group
- **Create AUTO Scaling Group**
 - Name
 - Launch Template
 - VPC
 - Availability zones (Private)
 - Attach a New/Existed Load Balancer
 - Internal
 - Public subnets
 - Scheme-Internal Facing
 - Target Group (New/Old)
 - Scaling
 - Desired(3) – because of selected availability zones
 - Min, Max
 - Target percentage to create
- **SETUP ROUTE53**
 - Create a Host zone for the domain
 - Create a record with the subdomain if required
 - Select the load balancer in the region add that cname of the loadbalance in the godaddy

YOU CAN SUCCESSFULLY ACCESS THE APPLICATION